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Thermodynamic assessments of the Ho–X (X: Cu, Mo, V) systems

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ABSTRACT

The thermodynamic assessments of the Ho–X (X: Cu, Mo, V) systems have been carried out by using the CALPHAD (Calculation of Phase Diagrams) method based on available experimental data including thermodynamic properties and phase equilibria. Gibbs free energies of the solution phases (liquid, hcp, fcc and bcc phase) were described by the subregular solution model with the Redlich–Kister formula, and those of the intermetallic compounds (αCu_5Ho, βCu_5Ho, Cu_9Ho_2, Cu_7Ho_2, Cu_2Ho and CuHo compound) in the Cu–Ho system were described by the sublattice model. Good agreement between the present calculated results and experimental data was obtained.

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1. Introduction

Copper base alloys are widely used in many fields because of their good combination of high thermal and electrical conductivities and high strength. Addition of rare earth elements into copper alloy combined with proper heat treatment may help to achieve both high strength and high electrical conductivity. On the other hand, holmium (Ho) has been used in ceramics, lasers and nuclear industry [1]. Moreover, there has been increasing interest by material scientists in the intermetallic compounds formed between rare earth metals and transition metals because of their potential physical and chemical characteristics [2,3].

The CALPHAD method, which is a powerful tool to cut down on cost and time during development of materials [4], effectively provides a clear guideline for materials design. In recent years, the present authors have developed a thermodynamic database of Cu-based alloys within the framework of CALPHAD method, which has been applied in the material design of Cu-based alloys [5,6]. Currently our group is also establishing a thermodynamic database of rare earth alloys, which is important for the design of alloys with additions of rare earth elements. In the present paper, as part of the above-mentioned two thermodynamic database, the thermodynamic assessments of the Ho–X (X: Cu, Mo, V) binary systems were carried out by means of the CALPHAD method.

2. Thermodynamic model

The information of stable solid phases and the models used in the Ho–X (X: Cu, Mo, V) systems [7] are listed in Table 1.

2.1. Solution phases

Gibbs free energies of the solution phases of (liquid and hcp, fcc and bcc phases) in an A–B system, corresponding to the Cu–Ho, Ho–Mo and Ho–V systems, were described by the subregular solution model and regular solution model with the Redlich–Kister formula [8]. The molar Gibbs free energy of each solution phase in the A–B system is given as follows:

$$G_m^{\phi} = \sum_{i=A,B} {}^0G_i^{\phi} x_i + RT \sum_{i=A,B} x_i \ln x_i + x_A x_B \sum_{m=0}^{n} {}^mL_{A,B}^{\phi} (x_A - x_B)^m, \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of pure component i taken from SGTE database [9]; x_i represents the mole fraction of the component i; R is the gas constant and T the absolute temperature; ${}^mL_{A,B}^{\phi}$ is the binary interaction parameter and is expressed as:

$${}^mL_{A,B}^{\phi} = a + bT, \quad (2)$$

where the values of a and b are evaluated in the present work based on the experimental data.

When m is equal to zero, the Eq. (1) can be used to describe the regular solution phases.

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Table 1
Stable solid phases and the models used in the Ho–X (X: Cu, Mo, V) systems.

System	Phase	Prototype	Strukturbericht designation	Modeling phase	Used models
Cu–Ho	(Cu)	Cu	A1	FCC	Subregular solution model
	αCu_5Ho	$AuBe_5$	$C15_b$	$(Cu)_5(Ho)$	Sublattice model
	βCu_5Ho	$CaCu_5$	$D2_d$	$(Cu)_5(Ho)$	Sublattice model
	Cu_9Ho_2			$(Cu)_9(Ho)_2$	Sublattice model
	Cu_7Ho_2			$(Cu)_7(Ho)_2$	Sublattice model
	Cu_2Ho	$CeCu_2$		$(Cu)_2(Ho)$	Sublattice model
	CuHo	CsCl	B2	(Cu) (Ho)	Sublattice model
	(Ho)	Mg	A3	HCP	Subregular solution model
Mo–Ho	(Mo)	W	A2	BCC	Subregular solution model
	(Ho)	Mg	A3	HCP	Subregular solution model
V–Ho	(V)	W	A2	BCC	Subregular solution model
	(Ho)	Mg	A3	HCP	Subregular solution model

2.2. Intermetallic compounds

The intermetallic compounds (αCu_5Ho, βCu_5Ho, Cu_9Ho_2, Cu_7Ho_2, Cu_2Ho and CuHo) in the Cu–Ho system were treated as stoichiometric compounds. The Gibbs free energy of the Cu_pHo_q compound is formulated as:

$$\Delta G_f^{Cu_pHo_q} = G_m^{Cu_pHo_q} - p^0 G_{Cu}^{fcc} - q^0 G_{Ho}^{hcp}, \quad (3)$$

where $\Delta G_f^{Cu_pHo_q}$ represents the Gibbs free energy of formation per mole of formula unit Cu_pHo_q, which is described as

$$\Delta G_f^{Cu_pHo_q} = a' + b'T, \quad (4)$$

where the values of a' and b' are evaluated in the present work.

3. Experimental information

3.1. Cu–Ho system

Wunderlin et al. [10,11] studied the solid solubility of Ho in the (Cu) fcc phase with metallographic methods, and determined the eutectic temperature in the Cu-rich region to be 864 °C. The crystal structures of the intermediate phases were investigated by various researchers, as follows: the Cu_5Ho compound is present in two allotropic modifications, namely αCu_5Ho and βCu_5Ho. The βCu_5Ho compound with the $CaCu_5$-type structure [12] is stable at elevated temperatures, and the αCu_5Ho compound with the $AuBe_5$-type structure [13] is the stable phase at lower temperatures. The $\alpha Cu_5Ho \leftrightarrow \beta Cu_5Ho$ transformation temperature was estimated by Gschneidner and Calderwood [14] to be 970 °C. The Cu_2Ho compound crystallizes with the orthorhombic $CeCu_2$-type structure [15,16] and the CuHo compound with the cubic CsCl structure [17,18]. However, the existence of the Cu_9Ho_2 and Cu_7Ho_2 phases is proposed solely on the basis of the presence of similar phases in the Cu–Gd [19], Cu–Dy [20] and Cu–Er [21] systems. Based on thermodynamic considerations and the systematics of copper–lanthanide systems [22], Subramanian and Laughlin [23] reviewed the phase diagram of the Cu–Ho system, as shown in Fig. 1.

Fitzner and Kleppa [24] measured the standard enthalpy of formation of the CuHo phase by high temperature direct synthesis calorimetry at 1200 ± 2 °C. In later study by Meschel and Kleppa [25], the standard enthalpies of formation of the Cu_5Ho and Cu_2Ho phases were measured using the same method at 1100 ± 2 °C. Besides, the measurement of the mixing enthalpy in Cu–Ho binary system at 1250 °C was carried out by Nikolaenko [26] with high temperature calorimetry.

3.2. Mo–Ho system

The Mo–Ho system has a monotectic-type phase diagram with very low terminal solubilities and there exist no intermetallic compounds. The phase diagram in the Mo–Ho system reviewed by Subramanian [27] is shown in Fig. 2. There exists a stable miscibility gap in the liquid phase at high temperature, and the temperatures of the monotectic reaction $L_2 \leftrightarrow L_1 + (Mo)$ and the eutectic reaction $L \leftrightarrow (Mo) + (Ho)$ are estimated to be 2517 °C and 1432 °C, respectively. Currently no thermodynamic data are available for the Mo–Ho system.

3.3. V–Ho system

Savitskii [28] reported the V–Ho phase diagram to have an extensive liquid miscibility gap with a monotectic reaction ($L_2 \leftrightarrow L_1 + (V)$) near the melting temperature of pure V (1910 °C) and a eutectic reaction ($L \leftrightarrow (V) + (Ho)$) below the melting temperature of pure Ho (1474 °C), which is closely analogous to other vanadium–lanthanide phase diagrams. No intermetallic compound was reported in the V–Ho system. Smith and Lee [29] reviewed the phase diagram in the V–Ho system, as showed in Fig. 3, and evaluated the interaction parameters in the V–Ho system. However, these parameters for describing the Gibbs free energies of phases are not consistent with the SGTE pure element database [9]. Thus, it is necessary to reassessment the phase diagram in the V–Ho system. Currently no thermodynamic data are available for the V–Ho system.

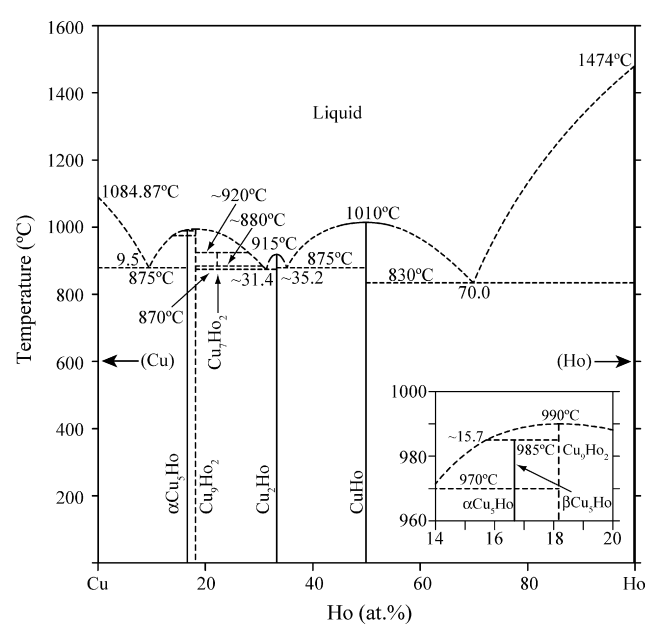

Fig. 1. The Cu–Ho phase diagram reviewed by Subramanian and Laughlin [23].

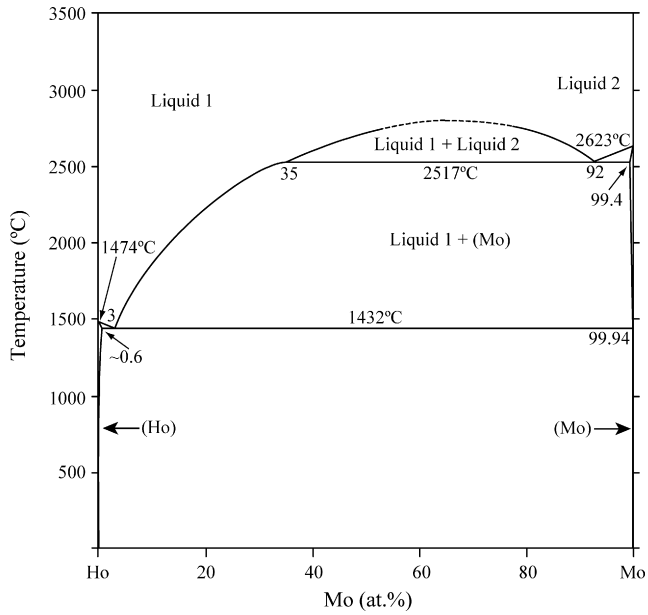

Fig. 2. The Mo–Ho phase diagram reviewed by Subramanian [27].

4. Results and discussion

The optimization was carried out using the PARROT module in the Thermo-Calc software [30], which can handle various kinds of experimental data. The phase diagram and thermochemical data were used as input to the program. Each piece of selected information was given a certain weight by personal judgment, and verified by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits. All the evaluated parameters of each phase in the Ho–X (X: Cu, Mo, V) binary systems are listed in Table 2.

4.1. Cu–Ho system

Comparison of the calculated Cu–Ho phase diagram with experimental information is illustrated in Fig. 4. The calculated phase diagram is in good agreement with the phase diagram reviewed

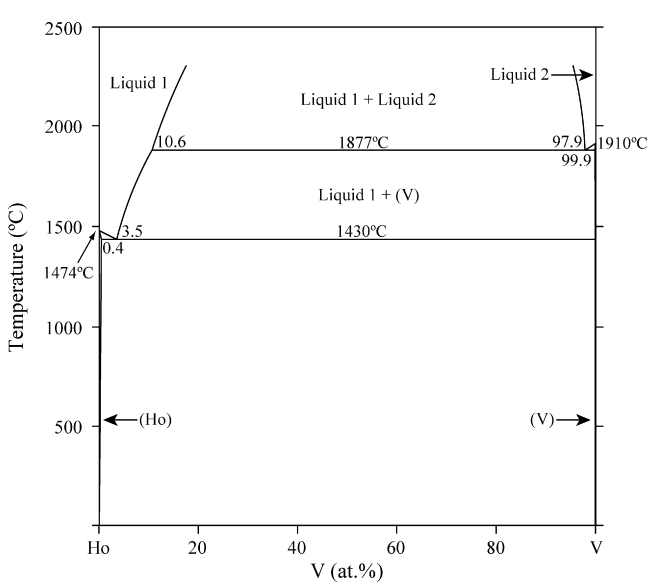

Fig. 3. The V–Ho phase diagram reviewed by Smith and Lee [29].

Table 2
Thermodynamic parameters in the Ho–X (X: Cu, Mo, V) systems.

System	Parameters in each phase (J / mol)
Cu–Ho	Liquid phase, formula (Cu, Ho): ${}^{0}L^{liq}_{Cu,Ho} = -76960 + 2.84T$ ${}^{1}L^{liq}_{Cu,Ho} = -26040 + 0.14T$ ${}^{2}L^{liq}_{Cu,Ho} = -16190 + 7.69T$
	FCC phase, formula (Cu, Ho): $G^{fcc}_{Ho} = G^{hcp}_{Ho} + 5000$ ${}^{0}L^{fcc}_{Cu,Ho} = 25000$
	HCP phase, formula (Cu, Ho): ${}^{0}L^{hcp}_{Cu,Ho} = 25000$
	αCu_5Ho phase, formula $(Cu)_5(Ho)_1$: $\Delta G_f^{\alpha Cu_5Ho} = -16612 + 0.4T$
	βCu_5Ho phase, formula $(Cu)_5(Ho)_1$: $\Delta G_f^{\beta Cu_5Ho} = -16438 + 0.26T$
	Cu_9Ho_2 phase, formula $(Cu)_9(Ho)_2$: $\Delta G_f^{Cu_9Ho_2} = -17620 + 0.4T$
	Cu_7Ho_2 phase, formula $(Cu)_7(Ho)_2$: $\Delta G_f^{Cu_7Ho_2} = -11445.2 - 6T$
	Cu_2Ho phase, formula $(Cu)_2(Ho)_1$: $\Delta G_f^{Cu_2Ho} = -23849 + 2.09T$
	CuHo phase, formula $(Cu)_1(Ho)_1$: $\Delta G_f^{CuHo} = -25690 + 2T$
Mo–Ho	Liquid phase, formula (Ho, Mo): ${}^{0}L^{liq}_{Ho,Mo} = 69000 - 7.3T$ ${}^{1}L^{liq}_{Ho,Mo} = -34000 + 6.5T$
	BCC phase, formula (Ho, Mo): ${}^{0}L^{bcc}_{Ho,Mo} = 116000$
	HCP phase, formula (Ho, Mo): ${}^{0}L^{hcp}_{Ho,Mo} = 55000$
V–Ho	Liquid phase, formula (Ho, V): ${}^{0}L^{liq}_{Ho,V} = 58686$ ${}^{1}L^{liq}_{Ho,V} = -10686$
	BCC phase, formula (Ho, V): ${}^{0}L^{bcc}_{Ho,V} = 98686$
	HCP phase, formula (Ho, V): ${}^{0}L^{hcp}_{Ho,V} = 66262$

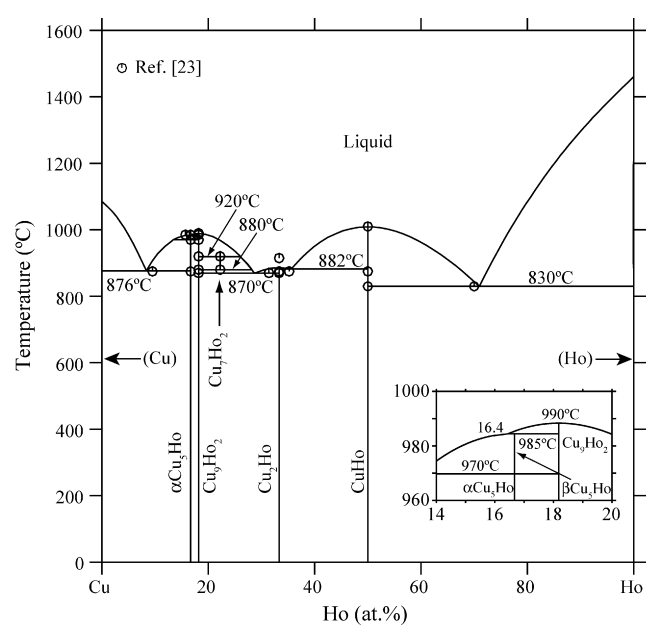

Fig. 4. The calculated Cu–Ho phase diagram with experimental data [23].

Table 3
Invariant reactions in the Cu–Ho system.

Reaction	Reaction type	Composition (at.% Ho)			T (°C)	Ref.
$L \leftrightarrow (Cu) + Cu_5Ho$	Eutectic		~0.02		864	[11]
		9.5	~0.02	16.7	875	[23]
		8.3	0	16.7	876	This work
$L + Cu_9Ho_2 \leftrightarrow \beta Cu_5Ho$	Peritectic	~15.7	18.2	16.7	985	[23]
		16.4	18.2	16.7	985	This work
$\alpha Cu_5Ho \leftrightarrow \beta Cu_5Ho$	Allotropic		16.7		970	[23]
			16.7		970	This work
$L \leftrightarrow Cu_9Ho_2$	Congruent		18.2		990	[23]
			18.2		990	This work
$L + Cu_9Ho_2 \leftrightarrow Cu_7Ho_2$	Peritectic		18.2	22.2	~920	[23]
		26.0	18.2	22.2	920	This work
$Cu_7Ho_2 \leftrightarrow L + Cu_9Ho_2$	Catatectic	22.2		18.2	~880	[23]
		22.2	28.2	18.2	880	This work
$L \leftrightarrow Cu_9Ho_2 + Cu_2Ho$	Eutectic	~31.4	18.2	33.3	870	[23]
		28.7	18.2	33.3	870	This work
$L \leftrightarrow Cu_2Ho$	Congruent		33.3		915	[23]
			33.3		890	This work
$L \leftrightarrow Cu_2Ho + CuHo$	Eutectic	~35.2	33.3	50.0	875	[23]
		35.6	33.3	50.0	882	This work
$L \leftrightarrow CuHo$	Congruent		50.0		1010	[23]
			50.0		1010	This work
$L \leftrightarrow CuHo + (Ho)$	Eutectic	70.0	50.0	~100	830	[23]
		70.9	50.0	100	830	This work

by Subramanian and Laughlin [23]. The invariant reactions in the Cu–Ho system are listed in Table 3, where good agreement between the calculated and experimental temperatures for those invariant reactions was obtained except for the congruent temperature of the Cu_2Ho phase, which can not be perfectly reproduced. Similar problem occurs in the thermodynamic description of the Cu–Dy [31], Cu–Er [32] and Cu–Tb [33] binary systems, where the congruent temperatures of the Cu_2RE (RE: Dy, Er and Tb) phases were not precisely reproduced. Therefore, further investigation seems to be needed to confirm the phase relations in this region.

The calculated standard enthalpies of formation of intermetallic phases are compared with experimental data in Fig. 5. The calculated results are reasonably consistent with the experimental results reported by Fitzner and Kleppa [24] and Meschel and Kleppa [25]. Moreover, the calculated enthalpy of mixing at 1250 °C is in excellent agreement with the experimental data, as shown in Fig. 6. The minimum values of mixing enthalpies of the liquid phase in Cu–RE systems (such as Cu–Tm [6], Cu–Dy [31], Cu–Er [32] and Cu–Yb [34]) correspond to around 40 at.% RE and the present calculated results can also fit this tendency.

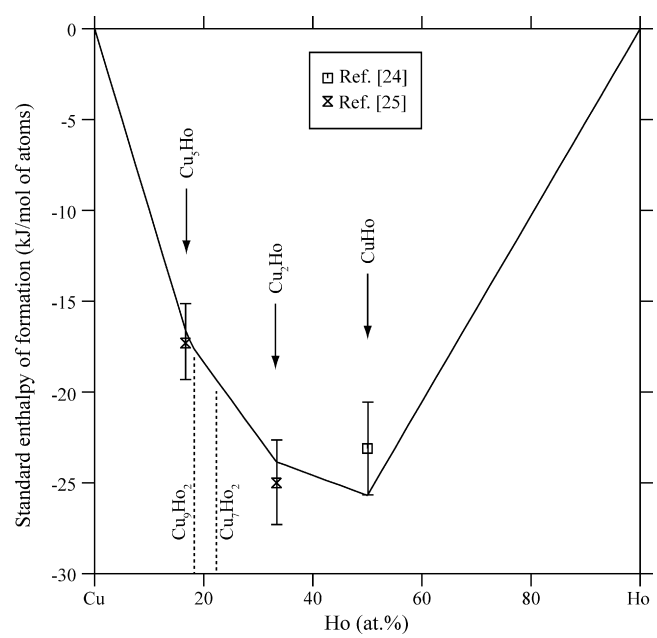

Fig. 5. The calculated standard enthalpies of formation of intermetallic compounds compared with experimental data [24,25].

4.2. Mo–Ho system

Since no experimental thermodynamic data in the Mo–Ho binary system is available, the optimization was carried out based on the Mo–Ho phase diagram reviewed by Subramanian [27] in the present work. The calculated phase diagram with experimental data is shown in Fig. 7. The invariant reactions in the Mo–Ho

Table 4
Invariant reactions in the Mo–Ho system.

Reaction	Reaction type	Composition (at.% Mo)			T (°C)	Ref.
$L_2 \leftrightarrow L_1 + (Mo)$	Monotectic	92	35	99.4	2517	[27]
		90.8	33.5	99.4	2517	This work
$L \leftrightarrow (Mo) + (Ho)$	Eutectic	3	99.94	~0.6	1432	[27]
		2.7	99.97	0.6	1432	This work

Table 5
Invariant reactions in the V–Ho system.

Reaction	Reaction type	Composition (at.% V)			T (°C)	Ref.
$L_2 \leftrightarrow L_1 + (V)$	Monotectic	97.9	10.6	99.9	1877	[29]
		97.6	8.8	99.7	1877	This work
$L \leftrightarrow (V) + (Ho)$	Eutectic	3.5	100	0.4	1430	[29]
		2.7	100	0.5	1430	This work

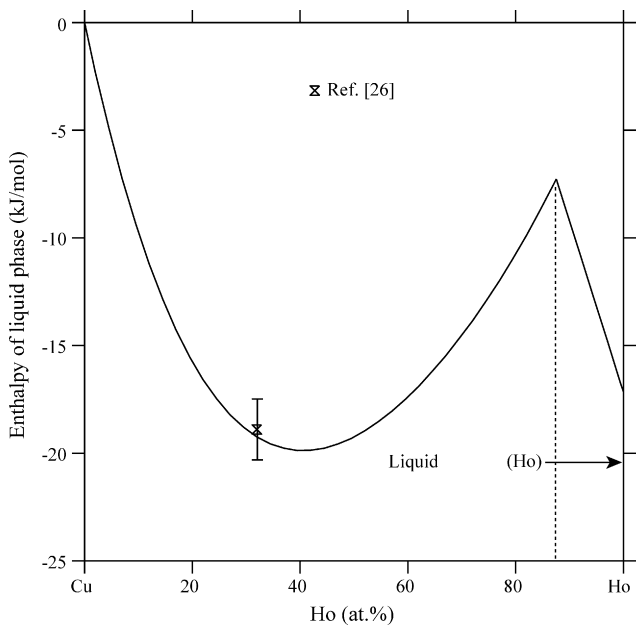

Fig. 6. The enthalpy for the alloying of liquid copper with 32 at.% Ho at 1250 °C with respect to the liquid elements [26].

system are summarized in Table 4. Based on the optimized parameters, the extrapolated miscibility gap in the liquid phase was also predicted in the Mo–Ho system, as shown in Fig. 7 by dotted lines. The calculated critical temperature and composition of the liquid miscibility gap are 2900 °C and 65 at.% Mo in the present work.

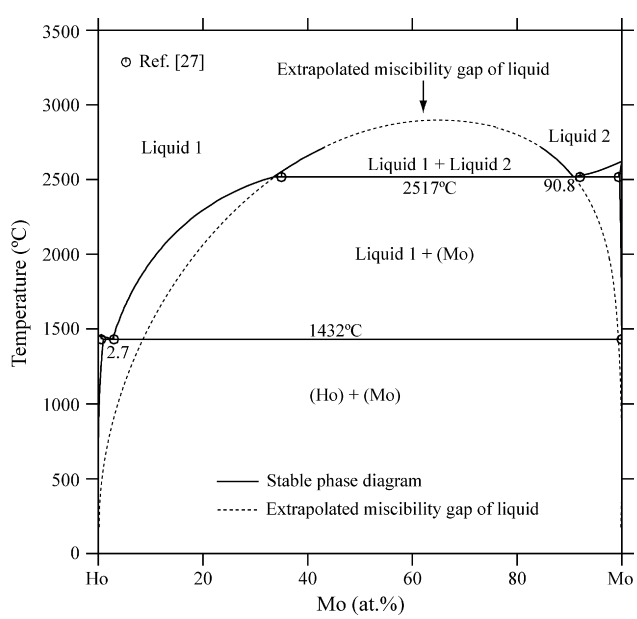

Fig. 7. The calculated Mo–Ho phase diagram with experimental data [27].

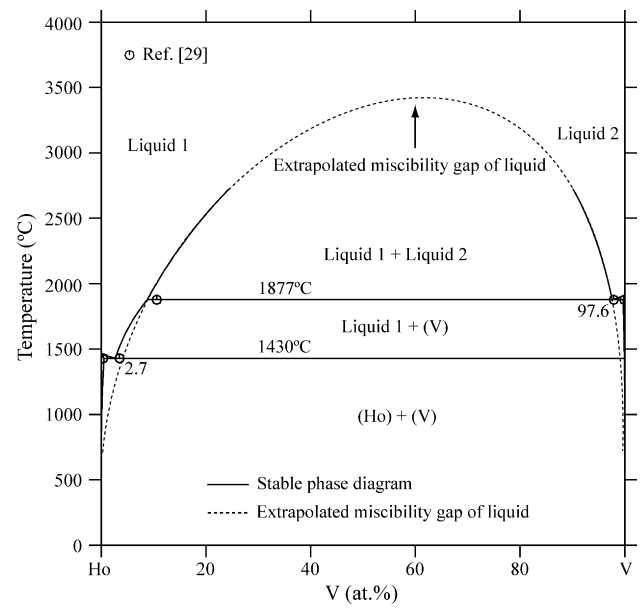

Fig. 8. The calculated V–Ho phase diagram with experimental data [29].

4.3. V–Ho system

The calculated phase diagram of the V–Ho system is shown in Fig. 8 together with the experimental information [29]. The invariant reactions in the V–Ho system are summarized in Table 5. The same way as in the Mo–Ho system, the extrapolated miscibility gap of the liquid phase in the V–Ho system was predicted on the basis of the optimized parameters in our work, as shown in Fig. 8 by dotted lines. It can be seen that the calculated critical temperature and composition of the liquid miscibility gap correspond to 3400 °C and 60 at.% V.

5. Summary

The Ho–X (X: Cu, Mo, V) binary systems were thermodynamically assessed based on available experimental data of phase diagrams and thermodynamic property. Good agreement between calculated results and experimental data has been reached and thermodynamic parameters for various phases in these three binary systems have been obtained.

Acknowledgements

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